Teaching notes

This activity is designed to give students a sense of the challenges facing ecologists when assessing the population of different plants.

This activity, with elements of a competitive game, is a useful alternative when a field trip or access to the school field is not possible.

The aim is to assess the accuracy of using quadrats as a technique for estimating the population of a plant species. The activity helps students understand the significance of positioning quadrats randomly and that populations can only be estimated as it is not possible in practice, to count all the individuals in an area.

However, this activity is not designed to be an accurate representation of how quadrats are used in the field. It is a fun activity to do on a wet day which will get students thinking.

The activity is based on the game ‘Battleships’.

Preparation

Equipment per pair of students:

- X2 sheets of blank grids
- stopwatch
- pencil

Each student should have a sheet of blank grids. The first blank grid on the sheet is to record the distribution of the plants in their survey area. The second blank grid is their record of the 'throws' they have made onto their partner's sample area.

The grids represent plots of land. You could assign a specific habitat to the plots e.g. saltmarsh, woodland, chalk downland, water meadow, heathland, sand dune.

A home learning task could be given as preparation for this activity. Students could find images of a plant species linked to one or more of the habitats listed above. They could write a description of the plant and it's habitat *in their own words*, using more than one source of information. They should write a list of references at the end.

Some suggestions of plants:

- Sea lavender *Limonium vulgare* – saltmarsh
- wood sorrel *Oxalis acetosella* – woodland
- wild thyme *Thymus praecox* – chalk downland
- creeping buttercup *Ranunculus repens* – water meadow
- tormentil *Potentilla erecta* – heathland
- eyebright *Euphrasia sp* – sand dune.

A follow-on home learning task could involve students finding out how ecologists working in the field, ensure that they have placed their quadrats randomly; or they could research other methods of surveying plants.

The activity

Explain to students that their grids represent areas of land and that they are going to investigate the most effective way of estimating the population of a certain species of plant, using a quadrat. For convenience, the 'quadrat' is the same size as one square on the grid.

From the beginning they will know the actual population size of the plant species. Explain that there are constraints on ecologists working in the field and that they do not have the resources to count all the individual plants in the area. How many random throws of the quadrat will give the best estimate?

Play

I. Decide for the class what percentage of the grid is to be occupied by the plant species. Each square is 4% of the whole, so choose multiples of 4, e.g. 32% = 8 squares.

II. Without showing others, students shade in the required number of squares on their first grid. The plants may be regularly spaced, clustered or concentrated in one small region.

III. Player 1 gives coordinates for a square, player 2 puts a cross through the square on their coloured grid and says if a 'hit' has been made. Player 1 marks the square on their empty grid to show if it was a hit or a miss.

Each turn represents one throw of the quadrat.

Play continues as Player 2 now gives the coordinates for a square.

In this initial game, play continues until all the plants have been found on one of the grids. Students need to time how long this takes.

IV. Class discussion

How long did it take them to find all the plants on one grid? How many 'throws' of the quadrat do they feel would have given a reliable estimate of the population?

What do they feel *is* a reliable figure? +/– 5% or 10%? What might it depend on?

Plants may not be evenly spread over the area of the grid. Will the reliability of their estimate compromised by the positioning of the plants?

V. Now they are ready to investigate further. Students can decide to limit the number of 'throws' to what they feel is a reasonable number. Alternatively, they may decide to give themselves a time limit for the game. These conditions represent the constraints of time and money on ecologists working in the field.

VI. Keeping the number of plants constant in each game, students play 3 games using the limited number of throws or time limit that they have decided upon.

What percentage population estimate did each game produce? What is the range of their results?

Further points for discussion

a) Point 2, '*without showing others, students colour the required number of squares of their first grid'* enables students to 'throw' their quadrats randomly.

What would happen if they had seen their opponent's grid with the plants drawn in?

What did happen when they scored a 'hit'? Did they concentrate their throws in that area?

How could they ensure that all their throws throughout the game, were random? Why is this important?

How could they investigate the effect of bias?

b) The 5 x 5 grid is small to allow more than one game to be carried out in a lesson. It represents a sample area of the larger habitat.

How would students use their results to provide an estimate of the whole habitat area? What would they need to know?
How accurate would this estimate be?
How could they test their ideas?

If time allows, pairs of students could come together into groups to carry out investigations to test some of these ideas.

Game 1

Number of throws

	A	B	C	D	E
1					
2					
3					
4					
5					

Time taken

	A	B	C	D	E
1					
2					
3					
4					
5					

Game 2

Number of throws

	A	B	C	D	E
1					
2					
3					
4					
5					

Time taken

	A	B	C	D	E
1					
2					
3					
4					
5					

Game 3

Number of throws

	A	B	C	D	E
1					
2					
3					
4					
5					

Time taken

	A	B	C	D	E
1					
2					
3					
4					
5					

Game 4

Number of throws

	A	B	C	D	E
1					
2					
3					
4					
5					

Time taken

	A	B	C	D	E
1					
2					
3					
4					
5					

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